

1 Plant Diversity

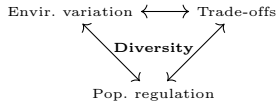
Diversity $\neq$ only species number: richness + evenness + genetic, functional, habitat diversity.

Diversity measure: species count/richness = # spp. (species); evenness = how equally individuals are distributed among species. Same richness but higher evenness $\Rightarrow$ higher diversity.

Diversity scales

α = diversity within a site (plot/habitat/location)
 β = diversity between sites (turnover across locations)
 γ = diversity in entire study area/region

Diversity framework



Env. variation different conditions select different traits.

Trade-offs no universal best strategy.

Population regulation (biotic interactions)

Density dependence: intraspecific competition limits population size; can promote coexistence.

Antagonistic interactions: interspecific competitors or consumers regulate population sizes.

Positive interactions: other species provide resources/services; can increase biodiversity.

2 Leaf Live Span

Leaf life span / leaf economics

LLS = leaf life span; varies widely in woody spp. (~2 mo–10 yr).

Key trade-off: fast C gain vs. long use of investment.

3 Approaches to study LLS variation: Comparative (compare species), theoretical/economical, mechanistic (chemical process level)

Fast leaves: short LLS; high SLA (specific leaf area), high leaf N, high photosynthesis/respiration.

Slow leaves: long LLS; tough epidermis/thick cell walls, low SLA/N, slow photosynthesis.

Leaf economics spectrum

LLS negatively related to $A_{\max}$ (max. rate of photosynthesis), SLA, leaf N.

SLA = onesided leaf area / leaf mass; LMA (leaf mass per area) = $1/\text{SLA}$.

Environmental effects

Cold climates: evergreen species $\Rightarrow$ longer LLS; deciduous species $\Rightarrow$ shorter LLS. $\Rightarrow$ **2 trends**

High resource availability: shorter LLS; moist/nutrient-rich sites favor faster leaf turnover.

Stressful/resource-poor sites: longer LLS, conservative strategy, avoid losses.

$$\text{NetC Gain} = \sum P_{\text{fav},i} F_{\text{dur}} + \sum P_{\text{unfav},i} U_{\text{dur}} - \text{CarbCost} - \text{Herb} - \text{StuckCarb}$$

Leaf payback / longevity

Minimal LLS: payback time = construction costs recovered; lifetime net C gain > 0 .

Optimal LLS: criterion-dependent: max net C gain per leaf/per time or efficient N/H₂O use.

Max. possible LLS: daily net C gain must stay > 0 ; if ≤ 0 , leaf dies soon.

Trade-off: allocation of Assimilatory tissue vs. Range of min/max LLS Assimilatory tissue $\equiv$ frac of tot leaf mass involved in photosynthetic activity, high SLA, high N

3 Plant Nutrition

All plants need the same essential mineral nutrients; all from soil/soil solution (N-fixers special case). Usually N and P most limiting.

N vs. P

N: taken up as NH_4^+ or NO_3^- ; mobile, easily leached/provided by microbes; atmospheric N only via N-fixation.

P: from weathered rock: less mobile, often bound to soil particles; soil P content $\neq$ P availability.

Old tropical soils often P-poor due to leaching (e.g. Amazon; partly replenished by Sahara dust).

Soil pH

pH affects nutrient availability, toxin concentrations and soil microbiota.

Low pH: possible Al toxicity; high pH: some nutrients poorly available.

NO_3^- uptake $\rightarrow$ OH^- exudation; NH_4^+ uptake $\rightarrow$ H^+ exudation.

Nutrient acquisition

Morphology: fine roots, root hairs, cluster roots. **max soil contact**

Chemistry: root exudates phosphatases to mobilize nutrients. 'Mining', up to 20% of tot C gain invested

Symbioses: N-fixing bacteria, mycorrhizae.

Traits for resorption

Resistant leaves, slow leaf senescence, thick roots with storage function, Rhizomes, Year-round growth

Photosynthetic N-use efficiency (PNUE)

$\text{PNUE} = C_{\text{assim}}/N_{\text{input}}$; N content = N input + N recycled.

High PNUE: high photosynthesis for given N, high SLA, high allocation to assimilatory tissue.

N residence time: long LLS + efficient N resorption before leaf fall.

N productivity – N residence time; LLS trade-off

N residence time increasing with LLS and N resorption.

LES trade-off: acquisitive leaves (high SLA, high N) have high short-term gain but short LLS; **conservative leaves** (LLS $\uparrow$; low SLA, low N, low SLA) have lower gain but long residence time.

Strategies range from fast turnover **high N prod** (early shedding, reinvest in new leaves) to steady-state withdrawal **high N res time** (gradual N resorption during aging).

N resorption is often high (about 50–90%; 50–98% for P), crucial in nutrient-poor or cold environments.

Nutrition and diversity

Different soil chemistry $\rightarrow$ different adapted species.

Within habitats: species differ in nutrient source, uptake timing and uptake depth $\rightarrow$ niche partitioning.

Nutrient enrichment $\rightarrow$ lost niche dimensionality/dominance $\rightarrow$ lower species richness.

4 Seasonal Life Cycle (Phenology)

Timing of life-cycle phases: germination, growth, flowering, seed maturation, dormancy.

Flower too early: frost damage, pollinator mismatch, slow development.

Flower too late: competition, herbivores, not enough time for seed maturation/development.

Climate warming often $\Rightarrow$ earlier spring development, but response limited by chilling requirements and photoperiod.

Forcing: warm temperatures accelerate development.

Chilling: winter cold required to release dormancy.

Phenology frequently implicated in spp. distribution limits. Trade-offs with other traits can sometimes explain limits to adaptation (and hence range limits).

5 Functional Traits and community assembly

Functional traits: Measurable plant properties affecting performance/function: survival, growth, reproduction, dispersal, competition, herbivory, pollination, responses to climate/soil/management.

Traits describe species/communities by function, not only species identity. traits $\equiv$ consistent strategies, can be plastic to some degree

Examples of func traits:

SLA; Leaf dry matter content (LDMC); *Life form and habit:* woody, deciduous/evergreen, creeping; *Life span* of the plant or of individual parts (leaves, roots); *Phenology:* seasonal timing of budbreak, flowering, dormancy; *Leaf anatomy:* width, thickness, specific leaf area, cell walls, hairs, spines *Seed traits:* seed size, mass, nutrient contents, longevity, dormancy, dispersal structures (pappus, wings, hooks) *Flower traits:* size, shape, odor, nectaries

Trait use

Functional types = species with similar traits/functions.

Single traits require clear definition: what measured? what function?

Community traits: *categorical* = proportion of spp. belonging to each category; *numeric* = community mean trait values, often weighted by the abundance of each species.

CSR Triangle, plant strategies based on traits: Competitor (*high height*), Stress-tolerators (*low SLA*), and Ruderals (small *seed size*, fast cycle; disturbance-adapted fast colonizer)

Grazing response

Resistance = avoid being grazed: toxins, hairs/spines, toughness, height.

Tolerance = survive/regenerate: protected meristems, uprooting resistance, vegetative propagation.

Escape = fecundity, colonize gaps: many seeds, seed size, dormancy, vegetative propagation.

Trait-based community assembly

Local species composition results from processes that add species (evolution, migration, dispersal) and filters that exclude species.

Environmental filter: unsuitable conditions exclude species directly or via environment-dependent biotic interactions. burning, mowing

Dispersal filter: limited dispersal prevents species from reaching suitable sites. seeding

Limiting similarity: similar neighboring species compete strongly; too similar species may not coexist locally.

6 Seed Dispersal

Seed dispersal in space

Dispersal: movement of seeds/propagules away from parent.

Dispersal kernel: probability distribution of seed distance from parent; tail = rare long-distance dispersal.

Modes: autochory = gravity/ballistic; allochory = wind (anemochory), water (hydrochory), animals (zoochory). Seed size/shape affect dispersal.

Why disperse? *Escape (Janzen Connell)* = avoid parent/kin competition/enemies (seed density high near parent, but recruitment often higher farther away); *Colonization* = reach favorable unpredictable sites; *Directed dispersal* = animals deposit seeds in favorable sites.

Dispersal affects community assembly, invasion, climate range shifts and global biodiversity patterns.

Seed dispersal in time

Seed bank: viable seeds stored in soil; composition can differ from above-ground vegetation.

Seed banks allow persistence in temporally variable environments and matter for restoration/invasion management.

Dormancy: viable seeds, spores, and buds fail to germinate under conditions favorable for vegetative growth; this determines the timing and quantity of germination.

Dormancy types: innate = seed physiology at release; enforced = missing environmental conditions for early seedling growth; induced = acquired after seed ripening.

Annual model: $N_{t+1}/N_t = (1 - g)s + gF$; g = germination fraction, s = annual seed-bank survival, F = seeds produced per germinant.

Seed bank forms if $g < 1$ and $s > 0$. Greatest population growth if $g = 1$ (no seed bank), but then $N_{t+1}/N_t = F$; if $F = 0$ in any year, population goes extinct.

7 Intraspecific Competition

Competition = reciprocal negative interaction; fitness reduced by shared limited resources (light/space, water/nutrients, mutualists).

Constant final yield: total biomass constant over wide density range; high density $\Rightarrow$ smaller individuals.

Density dependence: vital rates change with density; high density $\Rightarrow$ lower growth/fecundity/survival.

Self-thinning: as plants grow, weaker individuals die; density decreases, mean size increases.

–3/2 self-thinning law: $x = \log(\text{density})$, $y = \log(\text{mean individual mass})$; slope $\approx -3/2$.

Size hierarchies

Many small plants, few large plants; size distribution becomes increasingly skewed as plants grow. Small

plants are progressively lost through self-thinning.
Causes: relative growth rate differences, asymmetric competition, small-scale variation in density/resources, seed size/emergence time.

Asymmetric competition: larger individuals have disproportionately large effects on smaller ones, especially for light.

Implications: fitness differences increase; initial advantages matter a lot

Model: $N_{t+1} = \frac{\lambda N_t}{1 + \alpha N_t}$; N_t = pop. density now, λ = fecundity without neighbors, α = density-dependence strength.

8 Interspecific Competition

Competition: intensity vs. outcome

Individual level (competition intensity α): fitness/performance alone vs. with competitors → RCI (Relative Competition Index) or LRR (Log response ratio); indices simplify reality.

Population level (competition outcome): invasion, persistence, coexistence/exclusion.

Invasion criterion: species increases when rare while competitor is common; both invade ⇒ coexistence, one invades ⇒ exclusion.

2-species model

$N_i(t+1) = N_i(t)\lambda_i / (1 + \alpha_{ii}N_i + \alpha_{ij}N_j)$.
 λ_i = fecundity/rate increase without neighbors; α_{ii} = intra-specific effect; α_{ij} = inter-specific effect on species i .

Annual plants + seed bank

$N_{i,t+1} / N_{i,t} = (1 - g_i)s_i + g_i\lambda_i / (1 + \alpha_{ii}N_{i,t} + \alpha_{ij}N_{j,t})$.
 g = germination, s = seed-bank survival. Ability to recover from low density predicts persistence.

Good competitor

Insensitive to competition: small α_{ii}, α_{ij} ; strong effect on others: large α_{ji} .

Traits can predict competition effects/outcomes but only indirectly test mechanisms.

Coexistence possible via complementarity or facilitation in resource use.

R* theory: species with lowest resource requirement (lowest R*) wins competition for a limiting resource; it can reduce resource availability below competitors' persistence threshold.

Competition at individual level, decided at population level.

Coexistence: Niche Differentiation, Facilitation by other species, Environment changes. $\alpha_{ii} > \alpha_{ij}$

9 Pollination and Reproductive Success

Pollination needed for seed production; affects fitness, population dynamics and community structure.

Pollination modes

Wind pollination: widespread, massive pollen production, efficient in open habitats, costly and untargated.

Animal pollination: targeted pollen dispersal; floral displays/rewards; pollination syndromes (= flower traits matched to pollinator groups); specialization efficient but risky.

Breeding systems: Trade-off spectrum: reproductive assurance vs. inbreeding avoidance/genetic variation. Selfing/hermaphroditic flowers ⇒ reproductive assurance, but inbreeding risk.

Outcrossing or separate male/female flowers ⇒ less inbreeding + more genetic variation, but mate/pollinator dependent.

Pollen limitation: Plants produce fewer fruits/seeds than with adequate pollen receipt.

Test: pollen supplementation; if seed set increases ⇒ pollen limitation.

Pollen receipt is affected by conspecific density/distance, pollinator density/activity/diversity, pollen transport, and visit behavior.

Pollen limitation affects population growth mainly if seed limitation is strong and resource limitation weak.

Climate change: Different phenological shifts of plants and pollinators ⇒ temporal mismatch, stronger pollen limitation.

Different range shifts ⇒ spatial mismatch (latitude or elevation), especially risky for specialized interactions.

Invasions: Non-native plants can compete for pollinators or facilitate pollination. Low density may attract pollinators; high density may reduce native visitation/seed set.

10 Plant Soil Interactions

Plant Soil Interactions: Soil organisms affect plants; plants affect soil organisms.

Indirect effects: decomposers break down soil organic matter → N mineralization → plant-available nutrients; decomposers can also compete with plants for N.

Direct mutualists: N-fixing bacteria fix atmospheric N → NH_4^+ ; plants trade C for N and can sanction non-fixing cheaters (reducing their access to O_2). Mycorrhizae: plants trade C for mainly N/P and stress tolerance.

Direct enemies: soil pathogens/herbivores reduce plant performance; pathogen build-up can drive negative feedbacks/crop rotation/succession.

Plant effects on soil: root exudates feed microbes and can increase nutrient availability; litter quality controls decomposition rate.

Complex litter with high C:N or high condensed tannin:N ratio ⇒ slower soil organism activity/decomposition.

Plant-soil feedback (PSF): plants influence soil ecosystem, which affects subsequent plant performance. PSF can be net positive or negative; community consequences include succession, coexistence, rarity and invasion. Mutualisms are costly trading systems, not altruism; mycorrhizal networks can transfer resources/cues but rules remain unclear.

11 Plant-Herbivore Interactions 1

Plants are primary producers and cannot move ⇒ arms race with herbivores.

Herbivore pressure = herbivore abundance/diversity/development; **herbivory** = proportion of tissue consumed/damaged; fitness effects depend on plant response.

large/medium/small-specialized herbivores differ in feeding mode, mobility and specificity.

Plant strategies: Resistance, Tolerance and Escape

Resistance

Reduce/prevent attack.

Physical defenses: spines/thorns, trichomes, bark/cuticle, silica, leaf toughness, low N/H₂O.

Chemical defenses: phenolics/tannins, lignin, alkaloids, terpenes, glucosinolates.

Defense expression: constitutive = always expressed; induced = increased after damage/cues, often jasmonate-mediated.

Costs: physiological = resources for defenses, side-effects, phenology/stress sensitivity; ecological = deter mutualists, attract specialists, reduce tolerance/competitive ability.

Tolerance

Maintain fitness despite herbivore damage.

Mechanisms: storage reserves, rapid regrowth, protected meristems.

Measurement: compare fitness with vs. without herbivory.

Outcomes: undercompensation vs. overcompensation; compensation depends on favorable conditions/resources.

Escape

Avoid herbivory in space/time; leaves high/low, early life cycle, abundance fluctuations.

Defense syndromes

Plant defense strategies combine edibility/digestibility with toxicity/feeding barriers.

Tolerance/escape: fast growth in high-resource env.; high N, low toughness/chemical defenses, phenological escape.

Low nutritional quality: low H₂O/N, high toughness, digestibility-reducing compounds.

Nutrition + defense: high edibility (high N/H₂O/SLA, low toughness) balanced by defenses (trichomes, latex, toxins).

12 Plant-Herbivore Interactions 2

Herbivores affect plant competition, coexistence, diversity, and ecosystem nutrient distribution.

Specialist herbivores and coexistence

Specialist herbivores cause host-specific damage; common host species suffer stronger regulation →

rare species advantage → coexistence. Evidence/examples: Janzen–Connell effects and enemy release in plant invasions.

Janzen–Connell: conspecific seeds/seedlings near adult plants or at high conspecific density suffer higher enemy/pathogen mortality.

Mechanism: specialist enemies accumulate where host density is high, reducing survival close to parent plants.

conspecific = same spp.; heterospecific = different spp.

Enemy release & Invasion Success: non-native plants may lose specialist enemies → less regulation, greater competitive dominance.

Evolution of Increased Competitive Ability (EICA): loss of specialist natural enemies in new range → reallocate resources; less defense, more growth/competitive advantage possible.

Herbivores and diversity

Selective removal of dominant plants → releases resources (light, water, etc.) → and increases species richness.

Remove rare/palatable spp. or favor herbivore-tolerant/ resistant dominants → to lower spp. richness.

Plant diversity and herbivory

Associational susceptibility: greater herbivore damage with heterospecific neighbors (generalist herbivores, spill-over from other host spp.).

Associational resistance: reduced herbivore damage with heterospecific neighbors (dilution effect, physical isolation, chemical masking, more diverse natural enemies).

Plant diversity/composition affects herbivore pressure/damage; herbivore pressure often weaker in mixed stands than monocultures.

12 RR: Required Reading Summaries

PRR Debate (Adler vs. Fraser vs. Tredennick):

- **Q:** Does Richness peak at intermediate Productivity (HBM)? - **Adler ('11):** Standardized global data. *Result:* Productivity is a **poor predictor**. HBM rarely seen locally. - **Fraser ('15):** Targeted sampling of high-biomass sites. *Result:* Strong global evidence for HBM/Unimodal fit. - **Tredennick ('16):** Re-analysis of Fraser. *Method:* Added **Random Effects (Grid)** for independence. *Result:* HBM evidence disappears; supports Adler. Productivity ≠ driver.

Comita et al. ('10) - Rarity & NDD: - **Q:** Does **Conspecific NDD** explain species abundance? - **Method:** Seedling survival in 50-ha tropical plot. - **Result:** Rare species suffer **stronger NDD** (higher mortality near conspecifics) than common species. - **Conclusion:** Common species may dominate because they are **less sensitive** to specialized pests/pathogens.

Engelkes et al. ('08) - Enemy Release: - **Q:** Do climate range-expanders escape enemies? - **Method:** Feeding trials (locusts/aphids) on Native vs. Range-Expander vs. Exotic. - **Result:** Expanders and Exotics experience **lower herbivory/impact** than natives. - **Conclusion:** Range expansion is facilitated by **Enemy Release**; expanders leave specialized soil/leaf enemies behind.